

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. **Branch : Information Technology** **Semester : V**

Subject Code & Name: BTITPE503F Virtual Reality

Max Marks: 60 **Date: 02.02.2023** **Duration: 3 Hr.**

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Write a short note on illumination model in virtual reality?	CO3	6
B) Explain Painter's algorithm of hidden surface removal in virtual reality.	CO1	6
C) Explain shading algorithms in virtual reality?	CO1	6
Q.2 Solve Any Two of the following.		12
A) What is geometrical transformation? With example explain why do we use geometrical transformation in virtual reality?	CO2	6
B) What is virtual environment in virtual reality? Explain the different elements of virtual environment.	CO1	6
C) Explain the three types of virtual environment simulation.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) What is particle system? With the help of example explain it in detail.	CO1	6
B) What is inbetweening in virtual reality? Explain how it is implemented in virtual reality.	CO1	6
C) Explain the terms elastic collision and also explain direct impact and oblique impact of two particles in elastic collision.	CO1	6
Q.4 Solve Any Two of the following.		12
A) Explain the use of haptic communication in virtual reality.	CO1, CO2	6
B) Explain virtual reality suit in brief and also explain the applications of virtual reality suit.	CO2, CO3	6
C) For a company if you are asked to develop an application for digital marketing of their product how you will effectively use the virtual reality concepts that you have studied in the in your application.	CO2	6

Q. 5 Solve Any Two of the following.		12
A) Differentiate between augmented reality and virtual reality by using some applications?	CO1,CO3	6
B) Write a short note on applications of Virtual Reality.	CO3	6
C) Explain flight dynamics of an aircraft and how it is used for virtual reality application in aviation industry.	CO3	6

***** End *****